

Horimetrik: Value and Structure in a Positional Number System with Finite Factorial Horizons

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Abstract

We study a mixed-radix positional number system whose bases *decrease* towards the most significant position: a digit string $a_1 \cdots a_n$ with $0 \leq a_i \leq i$ denotes the value $\sum_i a_i (n+1)!/(i+1)!$. Such a system is sealed to the left; every number lives in a finite *horizon* of size $(n+1)!$, and a leading zero changes the value. Every element thereby carries two identities: its *value* and its *structure* (a polynomial with non-negative coefficients in the basis of falling factorials). We show: (1) The value- resp. structure-preserving embeddings of the finite horizons generate two different direct limit spaces, $\mathbb{N}_0$ and the positive structure semiring. (2) The space carries two mirror-image commutative semirings; in both, the same distinguished one acts as a projector onto the respective canonical copy. (3) A *side-fidelity theorem*: no commutative semiring arithmetic with value- resp. structure-carried operations can mix the two identities, independently of the choice of horizon rules; the obstruction is precisely the digit bound of the system. In addition we prove an asymptotic law for the commutator of the two compactifications, an extremal principle with exact growth laws of the structure horizon under addition, multiplication, finite difference, and composition (the composition closure of the positive semiring itself is part of the result), and document nine apparently unknown counting sequences. The text makes no claim of novelty beyond a documented search finding.

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1 The idea

Ordinary positional systems let the most significant position count farthest. This text studies the reverse convention. For fixed n , let a digit string $a_1 a_2 \cdots a_n$ with $0 \leq a_i \leq i$ denote the value

$$V_n(a) = \sum_{i=1}^n a_i \frac{(n+1)!}{(i+1)!};$$

the leftmost position is binary, yet carries the largest weight. Two consequences: the system is *sealed to the left* — below base 2 there is nothing, with n digits exactly $(n+1)!$ values are representable, and more room requires a new coordinate system. And *leading zeros are semantic*: $12 \mapsto 012 \mapsto 0012$ denotes 5, 6, 7 in turn.

Every number here thus carries two things: a value and a shape. The mathematics of this dual character is the subject of this text. The idea arose from the question of what happens if one reverses the exponent ordering of a number representation; the finiteness of the horizons is not a design goal but a necessity.

*Email: christian@buerckert.eu. Code, data, and an extended exposition (in German and English) are available at github.com/christianbuerckert/horimetrik. Licence: CC BY 4.0.

2 The Hori space

Definition 2.1. For $n \geq 0$ let $H_n = \{(a_1, \dots, a_n) : 0 \leq a_i \leq i\}$, so $|H_n| = (n+1)!$, with the value map V_n as above and the encoding $E_n = V_n^{-1}$. Further let $\mu(x) = \min\{n : x < (n+1)!\}$ be the *minimal horizon* of $x \in \mathbb{N}_0$.

Theorem 2.2. $V_n : H_n \rightarrow \{0, \dots, (n+1)! - 1\}$ is bijective.

Proof. Induction on n via $V_n(a) = (n+1)V_{n-1}(a_1, \dots, a_{n-1}) + a_n$ and $0 \leq a_n \leq n$: division with remainder by $n+1$ yields uniqueness; the range is $[0, (n+1)! - 1]$, and both sides have cardinality $(n+1)!$. $\square$

Definition 2.3. With $X^d = X(X-1)\cdots(X-d+1)$ let $\mathcal{S} = \{\sum_d c_d X^d : c_d \in \mathbb{N}_0\}$ and $\sigma(\sum c_d X^d) = \max_{c_d > 0} (d + c_d)$, $\sigma(0) = 0$. The *structure polynomial* of $a \in H_n$ is $P_a = \sum_{d=0}^{n-1} a_{n-d} X^d$; we have $V_n(a) = P_a(n+1)$, and $P \in \mathcal{S}$ is representable in H_n if and only if $\sigma(P) \leq n$. The *Hori space* is $\mathcal{H} = \{(n, P) : P \in \mathcal{S}, n \geq \sigma(P)\}$, equivalently $\{(n, x) : n \geq \mu(x)\}$.

$\mathcal{S}$ is closed under addition and multiplication (the product formula of the falling factorials has non-negative structure constants) and is therefore a commutative semiring.

3 First main result: two limit spaces

There are two natural ways to give an element more room: the *structure-preserving extension* $Z(n, P) = (n+1, P)$ (leading zero; the shape stays, the value becomes $P(n+2)$) and the *value-preserving lift* $L(n, P) = (n+1, E_{n+1}(P(n+1)))$ (the value stays, the shape is re-encoded). In the decimal system the two coincide; here they do not.

Theorem 3.1. As directed systems,

$$\varinjlim (H_n, Z_n) \cong \mathcal{S}, \quad \varinjlim (H_n, L_n) \cong \mathbb{N}_0.$$

Proof. Under Z , exactly those digit strings are identified that differ by leading zeros; each class carries exactly one structure polynomial, and every $P \in \mathcal{S}$ occurs (from $H_{\sigma(P)}$ onwards). Under L we have $L_{m,k} \circ L_{n,m} = L_{n,k}$ because $V \circ E = \text{id}$, and two elements are identified precisely when their values agree; every value occurs. $\square$

The same finite spaces thus generate, depending on the notion of consistency between the horizons, two different infinite worlds: the values $\mathbb{N}_0$ and the structures $\mathcal{S}$. Everything that follows concerns the relationship between these two worlds.

Remark 3.2 (third extension; Cantor coordinate). There is a third natural extension: *appending* a zero on the right, $R(a_1, \dots, a_n) = (a_1, \dots, a_n, 0)$. Because $V_n(a) = (n+1)! \cdot F(a)$ with the factorial fraction $F(a) = \sum_i a_i / (i+1)! \in [0, 1)$, the map R is *fraction-preserving* and scales the value exactly by $n+2$; the associated limit space is the set of finite Cantor factorial fractions. The obvious objection “essentially known since Cantor” thus applies exactly to this third, fraction-preserving world – not to the shape-preserving one (whose limit space is the structure semiring) and not to the theorems on the interplay proved below. Fraction, shape, and value are pairwise distinct invariants.

4 Second main result: dual projector semirings

The two *compactifications*¹ $K_V(n, x) = (\mu(x), x)$ (keep the value, smallest home) and $K_S(n, P) = (\sigma(P), P)$ (keep the shape, smallest home) are idempotent and do not commute; the smallest example is the digit string 010 in H_3 (value 4).

¹Not meant topologically: idempotent retractions onto the respective minimal horizon.

Theorem 4.1 (value side). *With*

$$(n, x) \oplus (m, y) = (\max(n, m, \mu(x+y)), x+y), \quad (n, x) \otimes (m, y) = (\mu(xy), xy)$$

$\mathcal{H}$ is a commutative semiring without a one; $(0, 0)$ is neutral and absorbing; the K_V -fixed points form a subsemiring isomorphic to $(\mathbb{N}_0, +, \cdot)$ with one $(1, 1)$, and for all h we have

$$h \otimes (1, 1) = K_V(h).$$

Proof. The core is the monotonicity of μ . Associativity of $\oplus$: both bracketings yield $\max(n, m, k, \mu(\text{partial sum}), \mu(x+y+z))$, and the inner μ -terms are absorbed by $\mu(x+y+z)$. Distributivity: $\max(\mu(xy), \mu(xz), \mu(xy+xz)) = \mu(x(y+z))$, again monotonicity. Neutrality of $(0, 0)$ follows from $n \geq \mu(x)$. An $\otimes$ -neutral element would have to force $\mu(x) = n$ for all elements, which fails at $(1, 0)$. Finally $(n, x) \otimes (1, 1) = (\mu(x), x) = K_V(n, x)$. $\square$

Lemma 4.2. $\sigma(P + Q) \geq \max(\sigma(P), \sigma(Q))$.

Proof. At the maximal position d^* of P we have $c_{d^*}(P+Q) \geq c_{d^*}(P) > 0$. $\square$

Theorem 4.3 (structure side, duality). *With*

$$(n, P) \boxplus (m, Q) = (\max(n, m, \sigma(P+Q)), P+Q), \quad (n, P) \boxtimes (m, Q) = (\sigma(PQ), PQ)$$

$\mathcal{H}$ is likewise a commutative semiring without a one; the K_S -fixed points form a subsemiring isomorphic to $\mathcal{S}$; K_S is a homomorphism for both operations, and for all h we have

$$h \boxtimes (1, 1) = K_S(h).$$

Proof. Verbatim the mirror image of Theorem 4.1, with Lemma 4.2 in place of the μ -monotonicity. The homomorphism property of K_S : for $\boxplus$, $\max(\sigma P, \sigma Q, \sigma(P+Q)) = \sigma(P+Q)$ by the lemma; for $\boxtimes$ both sides are $(\sigma(PQ), PQ)$. $\square$

Remark 4.4. It is *the same* Hori number $(1, 1)$ — the one in its smallest home — that on the value side carries K_V and on the structure side K_S as inner multiplication. The non-commuting of the compactifications is the non-commuting of two multiplications by one and the same element. Incidentally, the transformation monoid generated by K_S, K_V is finite: it has exactly six elements with the relation $K_V K_S K_V = K_S K_V$ (via a stability lemma: K_S preserves value-compactness) and five idempotents (proof in Appendix B).

5 Third main result: the side-fidelity theorem

Call an operation on $\mathcal{H}$ *value-carried* if the result value is the sum resp. product of the operand values (the result horizon is free, provided it is valid), and *structure-carried* if the result polynomial is the sum resp. product of the operand polynomials (horizon free $\geq \sigma$). Theorems 4.1 and 4.3 show: both *side-pure* combinations are law-abiding with suitable horizon rules. The mixed ones never are:

Theorem 5.1 (side fidelity). (A) *If $\oplus$ is value-carried and $\otimes$ structure-carried, then already left-distributivity is unsatisfiable for every choice of horizon rules.* (B) *If $\oplus$ is structure-carried and $\otimes$ value-carried, then left-distributivity and commutativity of $\otimes$ are jointly unsatisfiable.*

Proof. (A). Let $u = (1, 1)$ (structure 1) and $e_1 = u$, $e_m = e_{m-1} \oplus u$, so $\text{val}(e_m) = m$; the structure Q_m of e_m is some valid representation of m . Left-distributivity yields inductively

$$\text{val}(a \otimes e_m) = m \cdot \text{val}(a \otimes u). \quad (1)$$

Step 1: every Q_m is constant. For $c \geq 1$, $a_c = (c, c)$ has the constant structure c ; constants evaluate to themselves everywhere, so $\text{val}(a_c \otimes u) = c$. With (1) it follows that $c Q_m(y_c+1) = mc$, hence $Q_m(y_c+1) = m$, where $y_c \geq \sigma(c Q_m) = \max_d(d + c q_d) \geq c$ grows without bound. All evaluation points lie in the strictly monotone range of Q_m (arguments $\geq \sigma + 1 \geq \deg + 2$); for $\deg Q_m \geq 1$ there would be at most one such point. Hence Q_m is constant, and with $a = u$ in (1) it follows that $Q_m = m$.

Step 2: constants die. Let $g = (2, 3)$ with structure X and $r = \text{val}(g \otimes u) = t_g + 1 \geq 3$ fixed. By Step 1, $g \otimes e_m$ has the structure mX with $\sigma(mX) = m+1$, so $\text{val}(g \otimes e_m) = m(T+1) \geq m(m+2)$; by (1) this value is mr — a contradiction for $m > r - 2$.

(B). Now structures add and values multiply. Let $f_1 = u$, $f_m = f_{m-1} \oplus u$; then f_m has the constant structure m and the value m . Let A be the structure of $g \otimes u$ (a representation of 3). Inductively, $g \otimes f_m$ has the structure mA ; as an $\otimes$ -result of value $3m$, realized on its horizon T_m , validity (the digit bound) forces $T_m \geq \sigma(mA) \geq m$ and $A(T_m+1) = 3$. As in Step 1, A is constant, so $A = 3$. Now consider $f'_2 = g \oplus g$ with structure $2X$, horizon $H \geq \sigma(2X) = 3$, and value $2(H+1) \geq 8$. Left-distributivity and commutativity give

$$u \otimes (g \oplus g) = (g \otimes u) \oplus (g \otimes u),$$

on the right with structure $2 \cdot 3 = 6$, hence value 6; on the left stands the value $2(H+1) \geq 8$. Contradiction. $\square$

Remark 5.2. Both contradictions use the same property: $\sigma(m \cdot P)$ grows linearly in m , because the digit bound $c_d \leq n - d$ forces large coefficients into large horizons. The rigidity of the two identities is thus a direct consequence of the defining digit condition $0 \leq a_i \leq i$. The non-commutative mixed case of type (B) is likewise excluded under two-sided distributivity: right-distributivity mirrors the chain construction and replaces commutativity. And side fidelity is by now *three-sided*: the fraction identity (Cantor coordinate) carries no arithmetic of its own — its addition can exceed one and is never total — and cannot be interwoven with either of the two sides (value- $\oplus$ fails at the constant factorial quotient, structure- $\oplus$ at the half-coefficient obstruction $2A(Y) = Y^k$; proofs in Appendix C). There are exactly two computational worlds.

6 The depth law of the commutator

The defect $\delta(h) = \text{hor}(K_S K_V h) - \text{hor}(K_V K_S h)$ is unbounded on both sides, but extremely asymmetric: upwards, $+2$ first occurs at $h = (9, 720)$ ($720 = 6! = 10 \cdot 9 \cdot 8$ has in H_9 the structure X^3 ; secured exhaustively); the further thresholds are bounded only from above: $+3$ at the latest from $n = 14$, $+4$ at the latest from $n = 20$, $+5$ at the latest from $n = 26$. Downwards, the depth $-g$ is tied to the *shell gap*; the jump positions are

$$m_g = 3, 15, 84, 533, 3883, 31998, 294875 \quad (g = 1, \dots, 7).$$

Theorem 6.1. *With $c_g = (g+2)! \sum_{k \geq 1} k/(g+k+1)! = 1 + O(1/g)$ we have*

$$\frac{(g+2)!}{c_g} \leq m_g + 1 \leq (g+2)!, \quad \text{in particular} \quad \frac{m_g}{(g+2)!} \rightarrow 1.$$

Proof. Reindexing yields exactly $M_{m-g}(m+1)/m! = (m+1) \sum_{k=1}^{m-g} k/(g+k+1)!$, where $M_s = \sum_{d < s} (s-d)X^d$ is the maximal polynomial. The right-hand side grows strictly in m ; for $m+1 \geq (g+2)!$ the term $k=1$ already suffices, and for $m+1 < (g+2)!/c_g$ even the full series is too small. The bound $c_g \leq 1 + \sum_{k \geq 2} k(g+3)^{1-k}$ gives $c_g = 1 + O(1/g)$. $\square$

Theorem 6.2 (depth law, exact equality). *For every shell $m \geq 3$, $\min\{\delta(h) \mid \mu(\text{val}(h)) = m\} = -g(m)$ with $g(m) = m - s_{\min}(m)$, $s_{\min}(m) = \min\{s \mid M_s(m+1) \geq m!\}$; the terminal representation of $x^* = M_{s_{\min}}(m+1)$ is a witness.*

Proof. Bound. Let $h = (n, x)$ with $\mu(x) = m$, so $x \geq m!$. Left: $K_V(h) = (m, x)$, and for $P = \text{struct}(x, m)$ we have $P \preceq M_{\sigma(P)}$ coefficientwise, hence $x = P(m+1) \leq M_{\sigma(P)}(m+1)$; since $x \geq m!$ it follows that $\sigma(P) \geq s_{\min}$ and thus $\text{hor}(K_S K_V(h)) = \sigma(P) \geq s_{\min}$. Right: for $P_n = \text{struct}(x, n)$ the evaluation is monotone (nonnegative coefficients), hence $\text{val}(K_S(h)) = P_n(\sigma(P_n)+1) \leq P_n(n+1) = x < (m+1)!$ and thus $\text{hor}(K_V K_S(h)) \leq m$. Together $\delta(h) \geq s_{\min} - m = -g(m)$.

Attainment. We have $x^* = M_{s_{\min}}(m+1) \in [m!, (m+1)!]$ – the lower bound by the definition of $s_{\min}$, the upper since $M_{s_{\min}} \preceq M_m$ and $M_m(m+1) = (m+1)! - 1$. Let $h^* = (x^*, x^*)$ be the terminal representation. K_S fixes h^* (constant shape), hence $\text{hor}(K_V K_S(h^*)) = \mu(x^*) = m$. On the other hand, $M_{s_{\min}}$ is a valid digit sequence in horizon m ($c_d \leq s_{\min} - d \leq m - d$) with value x^* , hence by the bijection from Theorem 2.2 the digit sequence: $\text{struct}(x^*, m) = M_{s_{\min}}$, and $\text{hor}(K_S K_V(h^*)) = \sigma(M_{s_{\min}}) = s_{\min}$. Thus $\delta(h^*) = s_{\min} - m = -g(m)$. $\square$

Empirically, $(g+2)!/c_g$ even equals, to within ± 1 , the value m_g ; the prediction $m_7 = 294\,875$ derived from this was confirmed *exactly*, as were two earlier predictions of this law (-3 from shell ≈ 100 ; $m_6 = 31998$ against a predicted ≈ 31930).

7 The horizon calculus

For a k -ary operator T on the structure semiring $\mathcal{S}$ let

$$\Gamma_T(r_1, \dots, r_k) = \max\{\sigma(T(P_1, \dots, P_k)) : \sigma(P_i) \leq r_i\}$$

be the maximal structure horizon of the result. We write $P \preceq Q$ for the coefficientwise order in the falling factorial basis and $M_r(X) = \sum_{d \leq r} (r-d) X^d$ for the maximally filled polynomial with $\sigma(M_r) = r$.

Theorem 7.1 (extremal principle). *If T is coefficient-monotone in each argument — which holds for every operator composed of addition, multiplication, and Δ , since all structure constants are non-negative —, then*

$$\Gamma_T(r_1, \dots, r_k) = \sigma(T(M_{r_1}, \dots, M_{r_k})).$$

Proof. $\sigma(P) \leq r$ is equivalent to $P \preceq M_r$ (digit bound $c_d \leq r - d$), and σ is $\preceq$ -monotone; monotonicity of T yields the upper bound, $M_{r_i} \in \mathcal{F}_{r_i}$ the attainment. $\square$

Thus every basic operation possesses its own exact growth law: $\Gamma_+(r, s) = r + s$ (linear, sharp at constants), $\Gamma_\times(r, s) = \beta(r, s) = \sigma(M_r M_s)$ (essentially factorial, see below) — and for the finite difference:

Theorem 7.2 (difference overflow). *For $r \geq 2$ we have $\Gamma_\Delta(r) = \lfloor (r+1)^2/4 \rfloor - 1$, attained at the single-digit structure $P = (r-d) X^d$, $d = \lfloor (r+1)/2 \rfloor$.*

Proof. By Theorem 7.1, $\Gamma_\Delta(r) = \sigma(\Delta M_r)$ with coefficients $b_k = (k+1)(r-k-1)$, hence $\Gamma_\Delta(r) = \max_{0 \leq k \leq r-2} (k+1)(r-k) - 1$; the product of two numbers with sum $r+1$ is at most $\lfloor (r+1)^2/4 \rfloor$. (Confirmed exhaustively up to $r = 8$: 1, 3, 5, 8, 11, 15, 19.) $\square$

So Δ lowers the degree and raises the capacity quadratically: the degree measures dimension, σ measures capacity; a degree filtration does not see this effect.

The calculus also captures substitution:

Theorem 7.3 (composition closure). *$\mathcal{S}$ is closed under composition, and $\circ$ is coefficient-monotone in both arguments; in particular $\Gamma_\circ(r, s) = \sigma(M_r \circ M_s)$.*

Proof sketch. Because $P \circ Q = \sum_d c_d(P)(Q)^d$, it suffices that $(Q)^d \in \mathcal{S}$. Interpreting $Q(x)$ as the number of pairs consisting of a template of arity k (a_k of each kind) and an injection $[k] \hookrightarrow [x]$, one sees that $Q(x)^d$ counts the d -tuples of pairwise distinct such pairs. Classification by the union of images yields $Q(x)^d = \sum_m N_m \binom{x}{m}$; on the exhausting tuples, S_m acts *freely* by relabelling the ground set (a permutation that fixes every injection fixes the exhausted union pointwise), so $m! \mid N_m$ and $(Q)^d = \sum_m (N_m/m!) X^m \in \mathcal{S}$ — positivity and integrality in one stroke. Monotonicity in Q follows because larger template supplies only enlarge the tuple sets. (Confirmed exhaustively for $r, s \leq 4$; the complete proof is in Appendix A.) $\square$

The diagonal $\Gamma_\circ(r, r) = 1, 4, 23, 6541, 477149751, \dots$ grows tower-like and is the ninth OEIS-unknown sequence of the project; its asymptotics are open. The shift operator $E P(X) = P(X+1)$, as the simplest special case, costs exactly $\Gamma_E(r) = r + \lfloor r^2/4 \rfloor = \Gamma_\Delta(r+1)$ — shifting costs as much as differencing one horizon higher (Appendix A); this value sequence is the classical quarter-squares sequence (A024206), new is the law, not the sequence. The continuous analogue of the composition closure is classical (absolutely monotone functions, Faà di Bruno); Theorem 7.3 is the discrete lattice sharpening with the additional divisibility $m! \mid N_m$. Additively, $(\mathcal{F}_r)_r$ with $\mathcal{F}_r = \{P : \sigma(P) \leq r\}$, $|\mathcal{F}_r| = (r+1)!$, is a classically compatible filtration; multiplicatively it is non-linear with optimal growth law β . Via the Newton expansion ($c_d = \Delta^d P(0)/d!$), $\sigma(P) = \max_{\Delta^d P(0) > 0} (d + \Delta^d P(0)/d!)$ is a basis-free *capacity height*; it is not a bit complexity (σ of a constant c is c itself).

8 The counting width: a #P interpretation

Because $X^d = d! \binom{X}{d}$, every $P \in \mathcal{S}$ has non-negative integer coefficients in the binomial basis ($b_d = c_d d!$) and thus lies in the class of *binomial-good* polynomials, which are characterized as the (relativizing) polynomial #P closure operations (Ikenmeyer–Pak, FOCS 2022, arXiv:2204.13149). $\mathcal{S}$ is a *proper* subclass ($\binom{X}{2} \notin \mathcal{S}$); the $d!$ -divisibility corresponds to ordered tuples instead of unordered selections.

Theorem 8.1 (Hori #P theorem). *For $f \in \#P$ and $P \in \mathcal{S}$ we have $P \circ f \in \#P$.*

Proof. Guess $d \leq \deg P$, a variant $\ell \leq c_d$, and d witness candidates; accept upon d pairwise distinct valid witnesses. The number of accepting paths is $\sum_d c_d f(x)^d = P(f(x))$; since P is fixed, everything remains polynomial. $\square$

In this reading the Hori operations acquire a counting semantics: addition is disjoint union of counting constructions, multiplication couples constructions (the linearization coefficients $\binom{p}{j} \binom{q}{j} j!$ count the overlap patterns), Δ counts the structures enabled by one additional witness ($(n+1)^d - n^d = d n^{d-1}$), and $\sigma(P) = \max(d + c_d)$ measures the maximal combination of witness count and variant count of a building block (*counting width*; geometrically the smallest staircase above the coefficient profile). Each $\mathcal{F}_r$ is thus a finite family of exactly $(r+1)!$ #P closure operators, and the horizon calculus quantifies optimally the growth of the counting width under $+$, $\times$, Δ , and $\circ$. Demarcation: the binomial basis conjecture of Ikenmeyer–Pak (whose univariate version would imply $P \neq NP$) remains untouched; the Hori operators lie on the known, relativizing side. New is exclusively the quantitative hierarchy.

9 Nine counting sequences

The following sequences originate from Horimetrik and produced no hit in the OEIS (last checked on 14 July 2026; a negative finding is no proof of novelty). Submission is under way; extended data tables (jump positions up to $g = 20$, jumper, non-jumper, and fixed-point

sequences up to $n = 40$, interchange up to H_{12} , 78 single-digit terms) are available as b-files in the repository:

Sequence	First terms
$\beta(r, r)$, maximal structure height of products	1, 6, 22, 87, 407, 2473, 19527, ...
Defect $D(n)$ (one-step shell jumpers)	1, 8, 60, 360, 2520, 21168, 211680
two-sided fixed points $F(n) = n \cdot n! - D(n)$	4, 17, 88, 540, 3960, 32760
jump positions m_g of the depth law	3, 15, 84, 533, 3883, 31998, 294875
interchange maxima $\max \Omega$ on H_n	0, 1, 6, 43, 351, 2873, 26443, 270291
interchange zero count $\#\{\Omega = 0\}$	2, 5, 13, 23, 61, 111, 274, 564
non-jumpers $n! - D(n)$	5, 16, 60, 360, 2520, 19152
single-digit factorials $\{n : n! \text{ single-digit in } H_{\mu(n!)}\}$	1, 2, 5, 7, 11, 23, 29, 39, 59, 119
composition height $\Gamma_o(r, r)$	1, 4, 23, 6541, 477149751, ...

Here $\Omega(h) = \text{val}(LZh) - \text{val}(ZLh)$ is the defect between the two growth orders; the positivity $\Omega \geq 0$ is proved (multiplication lemma with a sharp regime boundary; additionally co-confirmed exhaustively up to H_{12} , i.e. over $6.2 \cdot 10^9$ elements), the equality cases are characterized and completely unfolded into digit conditions (two inequalities per level with growing excess), and the zero count obeys a counting formula derived from this (exact for all measured terms) with a proved upper bound $(n+1)^{n+1}/n!$ — indifference to the growth order is asymptotically extremely rare (proofs in Appendix D). For $\beta(r, r)$ the asymptotics are proved: $\ln(\beta(r, r)/r!) = 2\sqrt{r} + O(\log r)$ (reduction to one maximal term plus elementary bounds; Appendix E). By the shell-jump lemma, $D(n)$ counts exactly the elements that a single leading zero lifts into the next factorial shell; the corresponding digit formula runs over the eigenrepresentation of $n!$, whose single-digit cases are completely characterized ($n = (i+1)!/d - 1$, $1 \leq d \leq i$). All nine sequences will be submitted to the OEIS; the A-numbers will be added to this text once assigned, so that text and database reference each other.

10 Context and honesty

The building blocks are known mathematics: mixed-radix systems and factoradics, Lehmer codes, falling factorials and finite differences, direct limits, product semirings. The closest known neighbours of the zero semantics are abstract numeration systems; there, leading zeros change the value via the radix order, without a semiring theory being erected on top. What we have not found in this form is the constellation: finite factorial horizons with running evaluation, value/structure duality with two projectors to the same one, and a rigidity theorem that forbids mixing the identities.

The philosophical motivation — infinity as a staircase of finite horizons — is motivation, not argument. The robust version reads: infinite structures can be understood as systems of finite horizons, and *different notions of consistency between the horizons generate different global spaces* (Theorem 3.1).

The base space itself captures the situation most precisely. If one augments the digits by a leading zero, $e = (0, a_1, \dots, a_n)$, then $0 \leq e_j < j$: as a set, H_n is the space of inversion sequences of length $n+1$, that is, $H_n \cong S_{n+1}$. The structure horizon becomes the permutation statistic $\sigma(P_a) = n - \min_{e_j > 0}((j-1) - e_j)$ with exact distribution $|\{\sigma = r\}| = r \cdot r!$ (Appendix F); whether it corresponds to a classical statistic is open. Coordinatewise, each horizon becomes a distributive lattice (the *middle order* on permutations, arXiv:2405.08943), in which $\mathcal{F}_r$ is the principal ideal below M_r . The inhabitants are thus known permutation codes; new are the operations on them — the reversed valuation, the horizon transitions, the duality, the side fidelity, and the capacity calculus. That is the most favourable situation for a young theory: known base space, new theorems.

Several earlier conjectures of this project were refuted by our own computations (one bound conjecture fell twice to test ranges that were too small; one formula conjecture at its fourth data point). We regard this as a quality feature of the method, not as a weakness of the subject.

11 Open questions

(1) The *upper* thresholds of the depth law exactly (so far only “at the latest from” bounds; the lower side is completely proved). (2) Fine structure of the β -asymptotics (conjectured: $-\frac{3}{2} \ln r$ in the logarithm term). (3) General digit statistics of $n!$ in its own shell (the single-digit cases are completely characterized); from this, the distribution asymptotics of $D(n)$. (4) Classification of the *coupled* arithmetics within one side (beyond the product constructions). (5) A lower bound for the interchange zero count. (6) Γ_T for further operators (composition, shift) and the question whether the capacity height $\max_d(d + c_d)$ has already been studied elsewhere. (7) Formalization of the side-fidelity theorem (the falling factorials are already available in Lean 4/Mathlib).

Reproducibility. All enumerations and verifications of this text are available as a Python reference implementation with self-tests; every result carries in the working files the status proved/conjectured/refuted with a date. Source code, b-files, and the detailed book (German and English, with all proofs) are available at github.com/christianbuerckert/horimetrik.

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A Proofs for the horizon calculus

Complete proof of Theorem 7.3 (composition closure).

Proof. Since $P \circ Q = \sum_d c_d(P) (Q)^d$ with $(Q)^d = Q(Q-1) \cdots (Q-d+1)$, it suffices to show that $(Q)^d$ is an admissible shape. Write $Q = \sum_k a_k X^k$ and choose pairwise disjoint sets T_k with $|T_k| = a_k$ (*templates of arity k*). Let a Q -structure on $[x]$ be a pair (t, φ) consisting of a template $t \in T_k$ and an injection $\varphi: [k] \hookrightarrow [x]$; their number is $\sum_k a_k x^k = Q(x)$. Hence $Q(x)^d$ counts the d -tuples of pairwise distinct Q -structures on $[x]$.

Classify the tuples by the union $U \subseteq [x]$ of the images of all injections involved. The number N_m of tuples whose image union entirely exhausts a fixed m -set depends only on m , and for all $x \in \mathbb{N}_0$ the polynomial identity $Q(x)^d = \sum_m N_m \binom{x}{m}$ holds. The symmetric group S_m acts on the exhausting tuples by renaming the ground set, $\pi \cdot (t, \varphi) = (t, \pi \circ \varphi)$, and this action is *free*: if π fixes every member of the tuple, then π fixes every image pointwise, hence by exhaustion all of $[m]$, i.e. $\pi = \text{id}$. (Templates of arity 0 do no harm: they contribute nothing to the union.) Thus $m! \mid N_m$, and $(Q)^d = \sum_m (N_m/m!) X^m$ has coefficients in $\mathbb{N}_0$ – positivity and integrality in one stroke.

Monotonicity in P : an $\mathbb{N}_0$ -linear combination of the $(Q)^d$. Monotonicity in Q : $a_k \leq a'_k$ yields $T_k \subseteq T'_k$; every tuple of Q -structures is one of Q' -structures with the same image union, hence $N_m \leq N'_m$ term by term. (Closure and extremal principle confirmed computationally and exhaustively for $r, s \leq 4$, the closure additionally for 300 random pairs up to $\sigma = 8$.) $\square$

Theorem A.1 (shift overflow). *For $E P(X) = P(X+1)$, $\Gamma_E(r) = r + \lfloor r^2/4 \rfloor = \Gamma_\Delta(r+1)$.*

Proof. E is linear with nonnegative constants ($(X+1)^d = X^d + dX^{d-1}$), so the extremal principle applies: $\Gamma_E(r) = \sigma(E M_r)$ with coefficients $c_k = (r-k) + (k+1)(r-k-1)$ for $k \leq r-2$ and $c_{r-1} = 1$; thus $k + c_k = r + (k+1)(r-k-1)$, and the product of two numbers with sum r is at most $\lfloor r^2/4 \rfloor$. The second equality is the identity $r + \lfloor r^2/4 \rfloor = \lfloor (r+2)^2/4 \rfloor - 1$. $\square$

B Product principle and the compactification monoid

Theorem B.1 (product principle). *Every commutative semiring structure $(\uplus, \odot)$ on the excesses yields, via $(x, \varepsilon) \oplus^* (y, \delta) = (x+y, \varepsilon \uplus \delta)$, $(x, \varepsilon) \otimes^* (y, \delta) = (xy, \varepsilon \odot \delta)$, a law-abiding arithmetic on $\mathcal{H}$.*

Proof. The map $(n, x) \mapsto (x, n - \mu(x))$ is a bijection $\mathcal{H} \rightarrow \mathbb{N}_0 \times \mathbb{N}_0$ (the condition $n \geq \mu(x)$ is exactly $\varepsilon \geq 0$; the inverse is $(x, \varepsilon) \mapsto (\mu(x) + \varepsilon, x)$). The operations are defined componentwise, with $(\mathbb{N}_0, +, \cdot)$ in the value factor; a product of two commutative semirings satisfies all laws componentwise. The selectors are valid, since the resulting horizon is $\mu(\text{value}) + \text{excess} \geq \mu$. $\square$

Theorem B.2 (compactification monoid). *K_S preserves value-compactness. Consequently $K_V K_S K_V = K_S K_V$, and the transformation monoid generated by K_S, K_V has exactly six elements – id , K_S , K_V , $K_S K_V$, $K_V K_S$ and $K_S K_V K_S$ – of which all except $K_V K_S$ are idempotent.*

Proof. *Stability:* let (m, P) be value-compact; the element $(0, 0)$ is fixed by K_S , so let $m \geq 1$ and hence $P(m+1) \geq m!$. Let $s = \sigma(P)$; suppose we had $P(s+1) < s!$. For $s \leq t < m$, each term $c_d X^d$ grows from $t+1$ to $t+2$ by the factor $(t+2)/(t+2-d)$, hence, since $d \leq s-1 \leq t-1$, by at most $(t+2)/3$; thus $P(t+2) < t! (t+2)/3 \leq (t+1)!$, and induction yields $P(m+1) < m!$ – contradiction. *Relation:* for every h , $K_V(h)$ is value-compact, by stability so is $K_S(K_V(h))$, hence K_V fixes this element. *Six elements:* every alternating word of length ≥ 4 contains $K_V K_S K_V$ and reduces; squares reduce by idempotence – at most six. Pairwise distinctness by explicit separating witnesses (finite enumeration); in particular $h = (6, 6)$ separates: $K_V K_S(h) = (3, 6)$, but $K_S K_V K_S(h) = (2, 5)$. Hence $(K_V K_S)^2 = K_S K_V K_S \neq K_V K_S$: not idempotent; the idempotence of the remaining five is verified using the relation. $\square$

C Side fidelity: the noncommutative case and the fraction identity

Theorem C.1 (noncommutative case). *If $\oplus$ is structure-carried and $\otimes$ value-carried and both distributive laws hold, then the system is unsatisfiable – without commutativity, associativity, or neutral elements.*

Proof. Right distributivity yields inductively $P_{f_m \otimes a} = m \cdot P_{u \otimes a}$; since $f_m \otimes a$ has the value $m \cdot \text{val}(a)$ and the horizon $\geq \sigma(m P_{u \otimes a}) \geq m$ grows, $P_{u \otimes a}$ is constant equal to $\text{val}(a)$ (monotonicity argument as above). In particular $P_{u \otimes g} = 3$; the $g \oplus g$ finale then runs word for word as in the commutative case via left distributivity. $\square$

Theorem C.2 (trilogy). *Fraction-carried addition is never total; value-carried addition with fraction-carried multiplication, as well as structure-carried addition with fraction-carried multiplication, are unsatisfiable for every choice of selectors (left distributivity suffices).*

Proof. *Totality:* $F(u) + F(u) = \frac{1}{2} + \frac{1}{2} = 1 \notin [0, 1)$, and every element has fraction < 1 .

Value $\oplus$, fraction $\otimes$: chains e_m with $\text{val}(e_m) = m$, horizon $N_m \geq \mu(m)$, fraction $m/(N_m+1)!$. Distributivity gives $\text{val}(u \otimes e_m) = m w$ with $w = \text{val}(u \otimes u)$ – an integer and ≥ 1 , since the fraction of $u \otimes u$ is $\frac{1}{4} > 0$. On the other hand, with h_m the horizon of $u \otimes e_m$,

$\text{val}(u \otimes e_m) = \frac{1}{2} \cdot \frac{m}{(N_m+1)!} (h_m+1)!$, hence $(h_m+1)!/(N_m+1)! = 2w > 1$ constant – but a factorial quotient > 1 is at least $N_m+2 \rightarrow \infty$.

Structure $\oplus$, fraction $\otimes$: chains f_m with constant structure m , horizon $N_m \geq m$, fraction $m/(N_m+1)!$. Distributivity gives $P_{u \otimes f_m} = m A$ with $A = P_{u \otimes u}$, $\deg A =: d$; comparing fractions on the horizon T_m of $u \otimes f_m$ yields, after cancelling m ,

$$2 A(T_m+1) = (T_m+1)^{\frac{k_m}{2}}, \quad k_m = T_m - N_m \geq 1.$$

The product on the right is at least $(N_m+k_m/2)^{k_m/2}$, the left-hand side at most $C(N_m+k_m+1)^d$; since $N_m \rightarrow \infty$, $k_m \leq 2d+1$ for large m , some value k^* occurs infinitely often, and $2A(Y) = Y^{k^*}$ holds as a polynomial identity. Then A would have the coefficient $\frac{1}{2}$ – digits are integers. (General multipliers a : the same computation with $c = (n_a+1)!/v_a > 1$ instead of 2.) $\square$

D The interchange defect: positivity, sharpness, zero count

For $x < (n+1)!$ let $z_n(x) := P(n+2)$ with $P = \text{struct}(x, n)$ – the value after one zero extension. The *interchange defect* of an element $h = (n, x)$ is $\Omega(h) = z_n(x) - z_{n+1}(x)$: the value after “grow first, then lift value-preservingly” minus the value after “lift first, then grow”.

Lemma D.1 (recursion formula). *For $x = q(n+1) + r$, $0 \leq r \leq n$: $z_n(x) = r + (n+2) z_{n-1}(q)$.*

Proof. From the digit recursion it follows that $P_{n,x}(X) = r + X \cdot P_{n-1,q}(X-1)$, since the coefficient shift by one degree is, by $X^d = X \cdot (X-1)^{d-1}$, multiplication by X with a shift of argument. Evaluate at $X = n+2$. $\square$

Lemma D.2 (strict monotonicity and superadditivity). *$z_n(x) < z_n(x+1)$, consequently $z_n(y+s) \geq z_n(y) + s$.*

Proof. Induction on n ; base $n = 1$: $z_1(0) = 0 < 1 = z_1(1)$. Without a carry, z_n grows by exactly 1; with a carry ($r = n$), $z_n(x+1) - z_n(x) = (n+2)(z_{n-1}(q+1) - z_{n-1}(q)) - n \geq (n+2) - n = 2$. $\square$

Lemma D.3 (multiplication lemma). *For $c \geq \ell+2$ and $cy < (\ell+1)!$, $z_\ell(cy) \geq (c+1) z_\ell(y)$.*

Proof. Induction on ℓ ; the base cases are empty and trivial, respectively. Let $y = q(\ell+1) + r$ and $cr = s(\ell+1) + u$ with $0 \leq u \leq \ell$; then $cy = (cq+s)(\ell+1) + u$ and, by the recursion formula, $z_\ell(cy) = u + (\ell+2) z_{\ell-1}(cq+s)$. Superadditivity and the induction hypothesis (the regime is preserved: $c \geq \ell+2 > (\ell-1)+2$; validity: $cq < \ell!$) give $z_{\ell-1}(cq+s) \geq (c+1) z_{\ell-1}(q) + s$. It remains to show $u + (\ell+2)s \geq (c+1)r$: with $c = \ell+1+e$, $e \geq 1$, we have $u = er \bmod (\ell+1)$, and the inequality is equivalent to $r(c-\ell-1) = er \geq er \bmod (\ell+1) = u$ – true. $\square$

Theorem D.4 (interchange positivity). *$\Omega(h) \geq 0$ for all $h \in \mathcal{H}$.*

Proof. By the definition of Ω it must be shown that $z_{n+1}(x) \leq z_n(x)$ for $x < (n+1)!$. Let $x = q'(n+2) + r'$. Then

$$z_{n+1}(x) = r' + (n+3) z_n(q') \leq r' + z_n((n+2)q') \leq z_n(q'(n+2) + r') = z_n(x),$$

first by the multiplication lemma ($c = n+2$, exactly at the regime boundary), then by superadditivity. $\square$

Theorem D.5 (equality cases). *$\Omega(h) = 0$ holds if and only if both steps of this chain are sharp: the multiplication lemma with $c = n+2$ for the sharp quotient q' , i.e. $z_n((n+2)q') = (n+3) z_n(q')$, and superadditivity carry-free, i.e. $z_n((n+2)q' + r') = z_n((n+2)q') + r'$.*

Proof. $\Omega(h)$ is the sum of the two deviations $z_n((n+2)q') - (n+3)z_n(q')$ and $z_n((n+2)q' + r') - z_n((n+2)q') - r'$; both are nonnegative by the multiplication lemma and superadditivity respectively, so their sum is zero if and only if both are zero. $\square$

Theorem D.6 (digit unfolding of sharpness). *For $c \geq \ell + 2$, $cy < (\ell + 1)!$, $y = q(\ell + 1) + r$, $e = c - \ell - 1$ we have: $z_\ell(cy) = (c+1)z_\ell(y)$ if and only if $er \leq \ell$, $(cq \bmod \ell) + r \leq \ell - 1$, and the same condition holds recursively for $(\ell - 1, c, q)$ (termination: for $\ell = 0$ the condition is empty).*

Proof. The chain $z_\ell(cy) = u + (\ell + 2)z_{\ell-1}(cq + s)$ from the proof of the multiplication lemma is sharp throughout exactly then: the arithmetic estimate when $er \leq \ell$ (then $s = r$, $u = er$), superadditivity when the $s = r$ single steps starting from cq are carry-free – their units digit is $cq \bmod \ell$, the condition therefore $(cq \bmod \ell) + r \leq \ell - 1$ –, the third recursively by definition. Verified exhaustively for $\ell \leq 7$. $\square$

Theorem D.7 (zero-count bound). $\#\{\Omega = 0 \text{ on } H_n\} \leq (n+1)^{n+1}/n!$, so the proportion falls at least like $e^{-(1+o(1))n \ln n}$.

Proof. By the equality theorem, every such element is determined by (q', r') with sharp q' , and carry-freeness forces $r' \leq n$ (the units digit grows by one per step and must stay below its maximum n before each step). By the digit unfolding, sharpness forces at level ℓ the digit bound $r_\ell \leq \lfloor \ell/(n+1-\ell) \rfloor$, and $\lfloor \ell/j \rfloor + 1 \leq (n+1)/j$ with $j = n+1-\ell$ yields the product $(n+1)^n/n!$ for the q' ; together with the $n+1$ possibilities for r' the bound follows. $\square$

E Digit formula, single-digit factorials, and the β asymptotics

Theorem E.1 (digit formula for the non-jumpers). *Let $d = (d_1, \dots, d_n)$ be the H_n digit sequence of $n!$ and P the first position with maximal digit ($d_P = P$), or $P = n$ if none exists. Then*

$$N(n) = \sum_{p=1}^P \min(d_p, p) \cdot \frac{n!}{p!}.$$

Proof. $N(n)$ counts digit sequences with $a_i \leq i - 1$ and value $< n!$. We decompose by the first position p at which a differs from d . The prefix must agree with d – possible only as long as $d_j \leq j - 1$, i.e. up to and including the first maximal digit. At position p we need $a_p < d_p$ and $a_p \leq p - 1$: $\min(d_p, p)$ possibilities. The suffix is free within the caps: $\prod_{j>p} j = n!/p!$ possibilities, and every such suffix stays strictly below, because the full (uncapped) suffix space fills the interval $[0, w_p)$ with the place weight $w_p = (n+1)!/(p+1)!$ bijectively. Every non-jumper is counted exactly once. $\square$

Theorem E.2 (β reduction). $\ln \beta(r, r) = \ln \max_t \binom{r-1}{t}^2 (r-1-t)! + O(\log r)$.

Proof. Lower bound: the triple $(p, q, j) = (r-1, r-1, r-1-t)$ contributes to degree $r-1+t$ the summand $\binom{r-1}{t}^2 (r-1-t)!$, and all contributions are nonnegative. Upper bound: each contribution is at most $r^2 \binom{r-1}{j}^2 j!$, and per degree there are at most r^3 triples; hence $\beta \leq 2r + r^5 \max_t(\cdot)$. $\square$

Theorem E.3 (β asymptotics). $\ln(\beta(r, r)/r!) = 2\sqrt{r} + O(\log r)$.

Proof. By the reduction theorem it suffices to prove the statement for the maximal term $\binom{s}{t}^2 (s-t)!/r! = s!/(r t!^2 (s-t)!)$ with $s = r - 1$. From above: Stirling bounds give $\ln(s!/(t!^2 (s-t)!)) \leq -h(t) + O(\log s)$ with $h(t) = 2t \ln t - t + (s-t) \ln(s-t) - s \ln s$; concavity yields $h(t) \geq t(2 \ln t - \ln s - 2)$, and this function has its minimum $-2\sqrt{s}$ at $t = \sqrt{s}$. From below: $t = \lceil \sqrt{s} \rceil$ with $\binom{s}{t} \geq (s-t)^t/t!$ gives $t \ln((s-t)/t^2) + 2t - O(\log r) = 2\sqrt{s} - O(\log r)$. $\square$

Theorem E.4 (single-digit factorials). *$n!$ is a single digit in H_n if and only if $n = (i+1)!/d - 1$ with $1 \leq d \leq i$; the digit is d at position i , and per position there are exactly i cases.*

Proof. A single digit d at position i means $n! = d(n+1)!/(i+1)!$, hence $d = (i+1)!/(n+1) \leq i$. Conversely, every $d \leq i$ divides $(i+1)!$, and $n+1 = (i+1)!/d$ makes the equation exact; the position exists, since $n+1 = (i+1)!/d \geq (i+1)!/i \geq i+1$ implies $i \leq n$, and the unique digit expansion then consists of exactly this digit. d runs bijectively through $1, \dots, i$. $\square$

F The inversion-sequence statistics

Theorem F.1 (inversion-sequence statistics). *H_n is in bijection with the inversion sequences of length $n+1$ and with S_{n+1} . The structure horizon is $\sigma(P_a) = n - \min_{e_j > 0}((j-1) - e_j)$ for $a \neq 0$ (for $a = 0$, $\sigma = 0$) with $e = (0, a_1, \dots, a_n)$, and $|\{a \in H_n : \sigma(P_a) = r\}| = r \cdot r!$ for $1 \leq r \leq n$.*

Proof. Setting $e_1 = 0$ and $e_{i+1} = a_i$, we get $0 \leq e_j < j$ for all $j = 1, \dots, n+1$ – the defining condition of an inversion sequence; the map $\pi \mapsto (e_j)_j$ with $e_j = |\{i < j : \pi_i > \pi_j\}|$ is the classical bijection to S_{n+1} , and both sides have $(n+1)!$ elements. For the structure horizon: $c_d = a_{n-d} = e_{n-d+1}$, hence $d + c_d = (n+1 - j) + e_j$ with $j = n - d + 1$; maximizing $d + c_d$ over $c_d > 0$ is minimizing $(j-1) - e_j$ over $e_j > 0$, and $\max(d + c_d) = n - \min_{e_j > 0}((j-1) - e_j)$. Counting: $\sigma(P_a) \leq r$ means $a_i \leq r - (n - i)$ for all i with $a_i > 0$, equivalently $a \leq M$ coordinatewise for the maximal profile belonging to H_r ; the number of such a in H_n is $(r+1)!$ (the digits above position r are forced to be zero, below they are free up to the bound). Hence $|\{\sigma \leq r\}| = (r+1)!$ and $|\{\sigma = r\}| = (r+1)! - r! = r \cdot r!$. (Confirmed exhaustively up to $n = 6$, `experiments_einordnung.py`.) $\square$

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